

A Novel Mechanism to Fuse Various Sub-aspect Brain-computer Interface (BCI) Systems with PSO for Motor Imagery Task

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Abstract—In this study, we develop a novel multi-fusion brain-computer interface (BCI) system based on a fuzzy neural network (FNN) and information fusion approaches to cope with a classification task for identifying right/left hand motor imagery. In the proposed system, we utilize a filter bank and sub-band common spatial pattern (SBCSP) to extract features from raw EEG data. A self-organizing neural fuzzy inference network (SONFIN) is then applied for a recognition task. In order to improve the classification performance, we form a committee of networks and employ fuzzy integral (FI) to attain a joint decision. To further optimize the fusion approaches, a particle swarm optimization (PSO) algorithm is exploited to globally update parameters used in the fusion stage. In consequence, our experimental result shows that the proposed fuzzy fusion system possesses superior performance compared to other comparative models.

Keywords—brain-computer interface (BCI); motor imagery (MI); sub-band common spatial pattern (SBCSP); fuzzy neural network (FNN); fuzzy integral (FI); particle swarm optimization (PSO)

I. INTRODUCTION

For decades, plentiful numbers of brain-computer interfaces (BCIs) have been proposed for real application [1, 2]. Conventionally, BCI systems must be equipped with the ability to describe a user's intent even without any peripheral muscular activity involved [3]. To meet the requirement, recordings of Electroencephalography (EEG) [4], functional near-infrared spectroscopy (fNIRS) [5], magnetic resonance imaging (MRI) [6] and many others are possible solutions to build effective BCI systems that utilize only brain wave activities. However, among all alternatives, the use of EEG outperforms than the others since it features high temporal resolution and low cost equipment.

In general, EEG signals are categorized into five frequency bands, including delta (<4Hz), theta (4-7Hz), alpha (8-12Hz), beta (12-30Hz) and gamma (>30Hz) [7]. Scientists have found that motor imagery alters the sensorimotor rhythms (SMR) concerning brain oscillations in the somatosensory and the motor area. In particular, alpha and beta activity in sensorimotor area are important indicators for motor imagery, in which alpha band activity in the sensorimotor area is also called mu activity.

A BCI system is designed and developed to help people overcome various motor disabilities. To serve the purpose, a

BCI system must be able to measure and process signals to detect certain patterns and send precise control commands in real time. In general, the process will be divided into three phases, signal preprocessing, feature extraction and pattern recognition. In the preprocessing phase, input signals are first arranged in a uniform format, and any noise or outlier existing in the signal will be removed while preserving all relevant information. Feature extraction can be regarded as an extension of the preprocessing phase. While all relevant information is preserved in the preprocessing phase, only the most informative features would be extracted in the feature extraction phase. This step assists the system in two perspectives: 1) enhancing computational efficiency, and 2) enhancing classification performance. Finally in the third phase, classification algorithms are applied to identify certain patterns according to the extracted features to further translate primitive signals into specific commands.

The following sections are arranged as follows. In section II, the motor imagery experiment and the proposed system are introduced. In section III, results are presented and some issues are discussed. Finally in section IV, a brief conclusion is made and future works are suggested.

II. MATERIALS AND METHODS

Our proposed algorithm is divided into two phases; the first one consists of processing the raw EEG data, including band-pass filtering, resampling, epoch extraction, and artifact removal, as illustrated in Fig. 1. The first phase of the algorithm corresponds to the aforementioned preprocessing step of an ordinary BCI system. It ensures that the processed EEG data is arranged in the format suitable for the second phase calculation and that the noisy data is removed as much as possible. As depicted in Fig. 2, the second phase of our proposed algorithm is composed of four stages. During the first and second stage, we establish a filter bank to separate each individual frequency band, including delta, theta, alpha, beta and the full frequency band (1-30 Hz). Consequently, the sub-band common spatial pattern (SBCSP) is further employed to extract spatial patterns based on each relative sub-frequency band.

In the third stage, we apply a fuzzy neural network as the classification algorithm to identify different EEG patterns. Additionally, in order to enhance classification accuracy and stability, we divide EEG signals into five frequency bands



Figure 1. Raw EEG data will be processed through various steps in the first phase, including band-pass filtering, re-sampling, epoch extraction, and artifact removal, before it can be fed into the second phase of the proposed algorithm.

(e.g., delta, theta, alpha, beta and full frequency band), and apply fuzzy integral (FI) to gather decisions from each frequency band. An Information fusion scheme is applied in the fourth stage to fuse decisions from each individual classifier. Finally, Particle swarm optimization (PSO) is conducted to tune weights of each classifier during fusion stage.

A. Filterbank And Sub-Band Common Spatial Pattern (SBCSP)

Common spatial pattern (CSP) is one of widely used methods of feature extraction in the EEG requisition task. The distribution of severe overlapping data can be clearly separated after using the CSP, which is shown in Fig. 3. Although it is effective to manually find the best band for each subject, it is also time consuming and the result is hardly repeated by other research groups. Here, we use a filter bank to decompose the EEG signals into 4 sub-bands, which include delta, theta, alpha, and beta bands. Additionally, a cross-band frequency feature is preserved to capture the interaction among different bands. This method refers to sub-band CSP [8] which has been proven to achieve similar result as finding proper bands manually for each subject. That is, SBCSP achieves higher accuracy than CSP and can provide more efficient and consistent result without manually finding proper bands for each subject.

B. Self-Organizing Neural Fuzzy Inference Network

An FNN is a combination of artificial neural networks and fuzzy logic as depicted in Fig. 4. This results in a hybrid intelligent system that synergizes these two techniques by combining the human-like reasoning style of fuzzy systems with the learning and architecture of neural networks. Furthermore, aiming at improving performance, the self-

organizing neural fuzzy inference network (SONFIN) was developed in [9]. To be elaborate, a SONFIN can adaptively generate clusters according to input data during training phase rather than pre-assigning number of clusters and cluster variables compared to conventional FNNs. The architecture of a SONFIN is composed of five layers, which is shown in Fig. 5. The mathematical functions of each layer in the SONFIN are explicitly described as follows. In the following description, $u^{(l)}$ denotes the output of a node in the l^{th} layer.

Layer 1 (Input layer) - No computation is done in layer 1. Each node in this layer, which corresponds to one input variable, only transmits input values to the next layer directly.

$$u_i^{(1)} = x_i \quad (1)$$

Layer 2 (Fuzzification layer) - Each node in this layer calculates the output from layer 1 into a fuzzy set as fuzzification layer.

$$u_{ij}^{(2)} = \exp\left(-\frac{[u_i^{(1)} - m_{ij}]^2}{\sigma_{ij}^2}\right) \quad (2)$$

where m_{ij} and σ_{ij}^2 are the mean and variance of the Gaussian membership function, respectively, of the j_{th} term of the i_{th} input variable x_i .

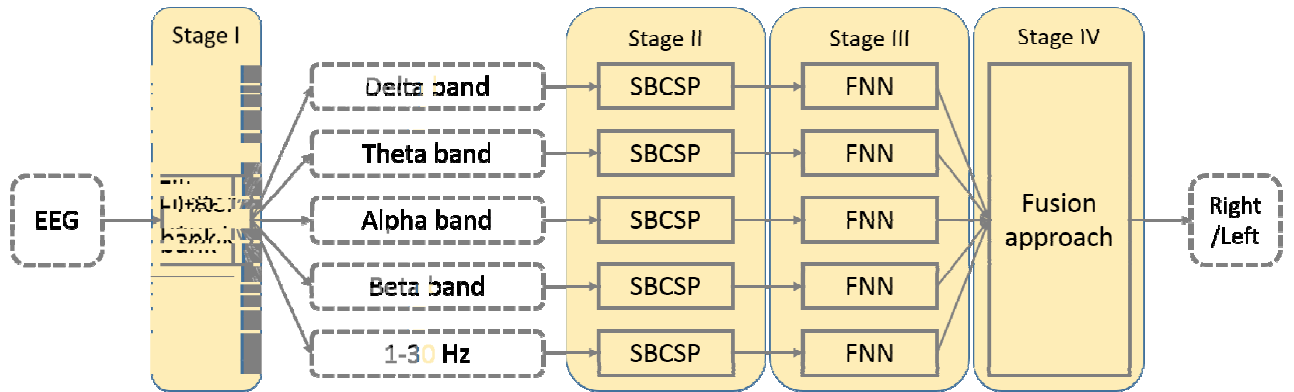


Figure 2. The second phase of our proposed algorithm is divided into four stages; filter bank and SBCSPs act as the feature extraction role, FNN classifiers identify different EEG pattern. In addition, a fusion approach is applied to fuse the result of the ensemble of FNNs.

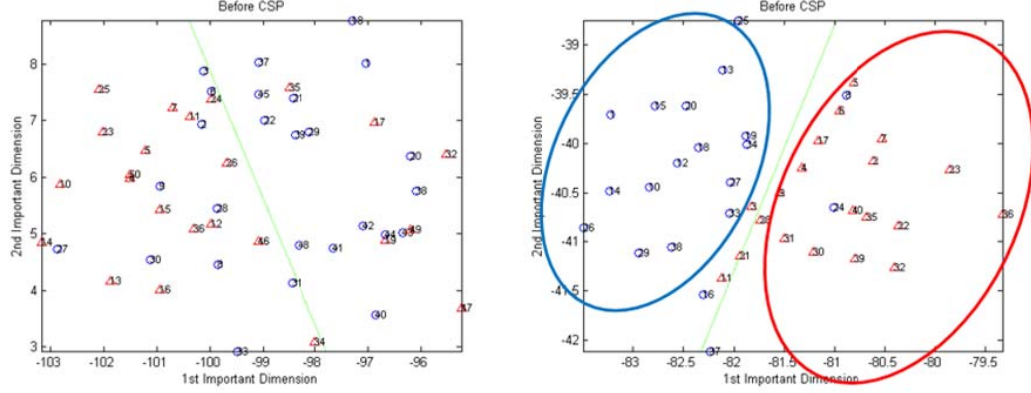


Figure 3. Scatter map before (left) and after (right) CSP projection.

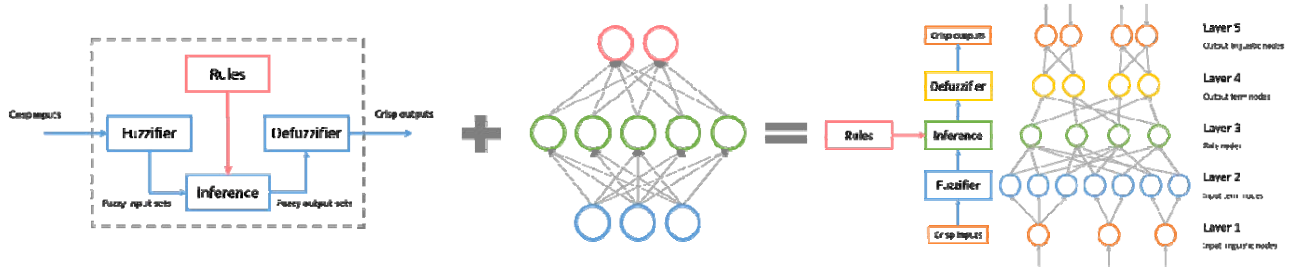


Figure 4. The structure of fuzzy neural network which arises from the combination of fuzzy inference system and conventional neural network.

Layer 3 (Rule layer) - The fuzzy AND operator is adopted to perform the precondition part of fuzzy rules. As a result, the output function of each rule node can be denoted by

$$u_j^{(3)} = \phi_j = \prod_j (u_{ij}^{(2)}) \quad (3)$$

where the $\prod_j u_{ij}^{(2)}$ of a rule node represents the firing strength of its corresponding rule.

Layer 4 (Consequent layer) - Nodes in layer 4 are called the consequent nodes. The function of consequent nodes forms the combination of input variables $\vec{X} = (x_1 \dots x_n)$.

$$u_j^{(4)} = \sum_{i=1}^n a_{ij} x_i(t) \quad (4)$$

Layer 5 (Output layer) - The output node in layer 5 integrates all of the actions from layers 4 and acts as a fuzzy defuzzification with

$$y = u_j^{(5)} = \frac{\sum_{j=1}^R \phi_j \left(\sum_{i=1}^n a_{ij} x_i \right)}{\sum_{j=1}^R \phi_j} \quad (5)$$

During the structure learning, a self-organizing fuzzy set generation algorithm is employed to construct fuzzy rules adaptively upon receiving training data. All of the fuzzy sets and rules in the SONFIN are generated adaptively, which not only automates the rule generation but also locates superior initial rule positions for subsequent parameter learning. During the parameter learning, all free parameters in antecedent and consequent part are learned through an on-line gradient descent algorithm using metrics derived from pattern sequences. The detailed training procedure of a SONFIN is illustrated in Fig. 6.

In our proposed system, we utilize five SONFINs to identify the right/left hand motor imagery according to extracted features from SBCSP. The outputs of each SONFINs are then combined and evaluated to attain a joint decision using fusion approaches.

C. Decision Fusion

In this section, three different types of fusion schemes, including weighted summation, Sugeno integral, and Choquet integral are introduced below. When the summation

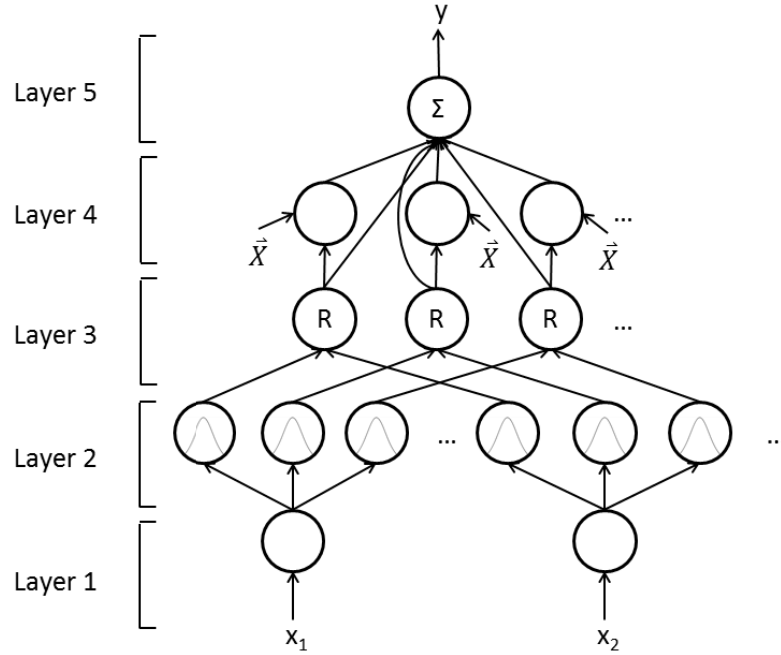


Figure 5. The architecture of the SONFIN.

is applied, we simply sum up the posterior probabilities of right movement and left movement from each individual classifier and select the class with the highest summation value as the system decision.

The Sugeno integral [10, 11] is a type of integral with respect to a fuzzy measure. The Sugeno integral over a crisp set A of a function h with respect to a fuzzy measure g is defined by

$$S_g(h) = \int_A h(x) \circ g = \sup_{\alpha \in [0,1]} [\min(\alpha, g(A \cap F_\alpha))] \quad (1)$$

where $F_\alpha = \{x | h(x) \geq \alpha\}$. When applying to a fusion problem, the function h is generally referred to the classifier output scores and the fuzzy measure g is the confidence or the weight we give to each classifier.

In addition, the Choquet integral [12], created by the French mathematician Gustave Choquet, is another type of integral with respect to a fuzzy measure. It is defined as:

$$C_g(h) = \sum_{i=1}^n [h(x_i) - h(x_{i-1})]g(A_i) \quad (2)$$

where $A_i = \{x_i, \dots, x_n\}$. As aforementioned, when applying into a fusion problem, h is referred to classifier outputs and g is the weight for each classifier.

D. Particle Swarm Optimization (PSO)

The PSO algorithm [13] is one of the well-known approaches of the swarm intelligence technology. It was developed by imitating movement of organisms in a bird flock or fish school. The PSO aims to optimize a model by iteratively trying to improve a candidate solution with regard to a given measure of quality. The PSO algorithm involves two critical steps:

1. Initialize a population of particles with random distribution within the desired range of the search space.
2. Update the particle position and velocity:

$$\begin{aligned} v_{id} &\leftarrow \omega v_{id} + \phi_p r_p (p_{id} - x_{id}) + \phi_g r_g (g_d - x_{id}) \\ x_i &\leftarrow x_i + v_i \end{aligned} \quad (3)$$

This procedure provides us with optimal weights for each information source in order to obtain global optimized output during the training phase.

E. Experimental Design

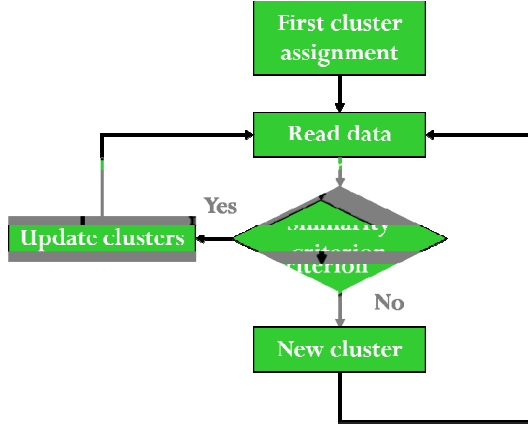


Figure 6. The procedure to determine the number of fuzzy rules (clusters) in the SONFIN structure.

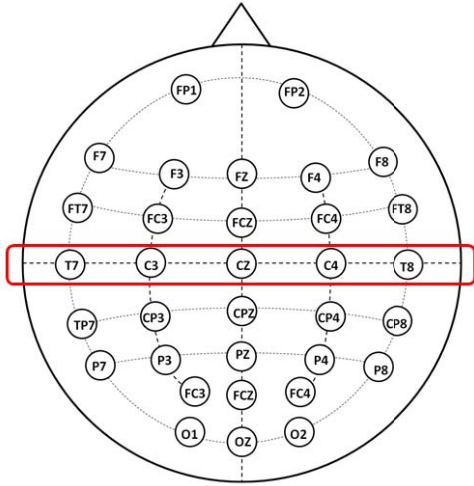


Figure 7. The electrodes are placed across the sensorimotor area, but not necessarily identical to the position in the 10-20 system.

In this study, we recruited four subjects and record brain activities during a motor imagery (MI) [14] task. Each subject was required to sit on a comfortable chair and place his/her hands on a table. In the beginning of each trial, the screen was kept blank for 2 seconds. After that, a cross was displayed on the center of the screen for 2 seconds and then an arrow point either to left or to right would randomly appear on the screen for 10 seconds. When the subject saw the arrow pointing to left, he was asked to do the imagery of left hand movement and vice versa. After the imagery, a picture would be displayed on the screen for a period between 7 to 10 seconds, which are randomly assigned to help subjects relax between two consecutive trials. Each scenario consists of 40 trials and there are totally 4 scenarios in each experiment. All EEG data were recorded using a portable 4-channel EEG recording device with dry spring-loaded sensor [15]. The EEG data were collected from the sensorimotor area [16] as shown in Fig. 7. The sampling rate is at 512Hz and later down-sam

TABLE I. CLASSIFICATION RESULT FOR BASE CLASSIFIERS AND DIFFERENCE FUSION TECHNIQUES.

	Area Under Curve of ROC	
	SONFIN_delta	0.965±0.030
Component SONFINs	SONFIN_theta	0.962±0.042
	SONFIN_alpha	0.961±0.039
	SONFIN_beta	0.946±0.055
	SONFIN_all band	0.957±0.043
Fusion Techniques	Summation	0.995±0.008
	Sugeno	0.990±0.013
	Choquet	0.990±0.002

TABLE II. CLASSIFICATION RESULT FOR SUGENO AND CHOQUET INTEGRAL AFTER PSO TRAINING.

Fusion Techniques	Area Under Curve of ROC	Before PSO	After PSO
	Sugeno	0.990±0.013	0.995±0.009
	Choquet	0.990±0.002	0.999±0.001

The collected data were then preprocessed in the first phase of our proposed system. And afterwards, in the second phase of the system, filter bank, SBCSP are applied to extract features and five base SONFINs identify the specific pattern. Weighted summation, Sugeno integral, and Choquet integral are applied as fusion techniques to fuse the five base classifiers, delta, theta, alpha, beta and all-band SONFINs. Initially, the weight of each classifier in Sugeno and Choquet integral was equally set to 1/5, which led to the result in Table I. Afterwards, the PSO algorithm would be applied to update these weights and formed the result in Table II.

III. RESULTS AND DISCUSSIONS

To compare the system performance in different fusion approaches, the receiver operating characteristic (ROC) curve is employed in this study, which is a fundamental tool for diagnostic test evaluation. The area under the ROC curve (AUC) is widely recognized as the measure of the performance. As shown in Table I, the average AUC value of individual classifier, including SONFIN_delta, SONFIN_theta, SONFIN_alpha, SONFIN_beta, SONFIN_all band, is 0.958, and the average AUC value after employing fusion technique reaches 0.992. This indicates that every fusion technique outperforms single classifier, and the proposed fusion architecture exhibits not only a higher AUC value but also a lower standard deviation. In comparison with the three fusion techniques, summation and Choquet integral are superior to Sugeno integral. Furthermore, we applied PSO to update the weights for classifiers in Sugeno and Choquet integral. As shown in Table II, with the Sugeno integral, we gain an improvement from 0.990±0.013 to 0.995±0.009, and an improvement from

0.990±0.002 to 0.999±0.001 with the Choquet integral. The results show that after tuning the weights with PSO, both the fuzzy integrals obtain a significant improvement. Not only the AUC is improved but also the standard deviation is reduced, suggesting that the system is performing at higher accuracy and better stability.

IV. CONCLUSIONS AND FUTURE WORK

In this study, we develop a novel BCI system to fuse sub-aspect information with the PSO algorithm for a MI recognition task. By applying the SBCSP, the SONFIN and the fusion approach, the proposed system is able to attain reliable performance. The highest AUC value is obtained when we adopt Choquet integral at the fusion level and exploit PSO algorithm to update the weights. The result suggests that the integration of a portable EEG collecting device and the proposed algorithm can serve as a practical BCI system in real-world scenarios. To better improve the BCI system used in real-life, we aim at developing a more advanced algorithm that is capable of classifying multiple-class motor imagery task and enables our system to execute more complex and diverse cognitive activities.

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